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Research on Optimal Control of Wet Desulfurization Slurry Feeding System

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abstract The wet flue gas desulfurization (WFGD) system is the core facility for controlling SO_2 $emissions$ in $coal-fired$ $power$ $plants$, $where$ the $control$ $accuracy$ of the $limestone$ $slurry$ $feeding$ $system$ $directly$ $affects$ the $desulfurization$ ef $interference$ $ability$, and $inability$ to $balance$ $multi-objective$ $optimization$ $when$ $facing$ the $strong$ $nonlinearity$, $large$ $time$ de

First, by analyzing the mechanism of the wet desulfurization process, a dynamic relationship model among the SO_2 $concentration$ at the $inlet$ of the $desulfurization$ $tower$, $flue$ gas $flow$ $rate$, $slurry$ pH $value$, and $slurry$ $feeding$ $objective$ $reward$ $function$ of the $system$ are $determined.$ $Second$, a $desulfurization$ $slurry$ $feeding$ $controller$ is $designed$ $using$ $time$ $operating$ $state$, $achieving$ the $collaborative$ $optimization$ of $desulfurization$ $efficiency$ and $slurry$ $consumption.$